EXERCICE 1

Give the monotonicity (increasing or decreasing) of each of the following sequences :

1. For all natural numbers n, $u_n = 7n - 12$.

2. $\forall n \in \mathbb{N}$, $v_n = \left(\frac{2}{3}\right)^n$.

3. $w_0 = 200$ and, for all natural numbers n :

$$w_{n+1} = -8 + w_n - n^2.$$

4. $t_0 = 4$ and, for all natural numbers n :

$$t_{n+1} = t_n^2 - 5t_n + 9.$$